WORKSHEET - NAPLAN Year 9

Algebra
- **Basic Algebra**

Number
- **Basic Concepts and Calculations**
- **Fractions & Decimals**
- **Percentages**
- **Word Problems**

Teacher: Brad Summers

Exam Equivalent Time: 40 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Algebra, NAP-14484

$5 \times ? + 15 = 95$

What value would make this number sentence correct?

10 14 16 20

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2. Number, NAP-85047

A return trip from Brodie's house to the beach is 5.78 kilometres.

How far does Brodie travel if he does this 14 times?

8.09 km 23.12 km 42.12 km 69.36 km 80.92 km

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3. Number, NAP-08568

In Australia, 288 701 children were enrolled in kindergarten in 2013.

Of these children, 150 125 were boys.

How many girls were enrolled in kindergarten in 2013?

122 896   138 576   288 701   438 826

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4. Number, NAP-66181

A butcher sells sausages in bags of four.

For a party, 20 sausages are needed.

How many bags of sausages are needed?

4 5 25 80

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5. Algebra, NAP-90964

If $p = 7$, what is the value of $2p$?

14 27 49 98

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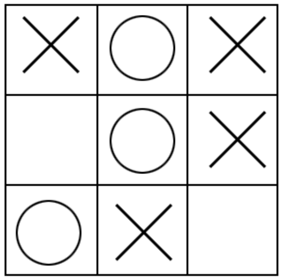
6. Number, NAP-61527

The image below shows a game of noughts and crosses in progress.



What fraction of the available boxes have been filled in?

$\frac{1}{2}$ $\frac{4}{7}$ $\frac{5}{7}$ $\frac{7}{9}$

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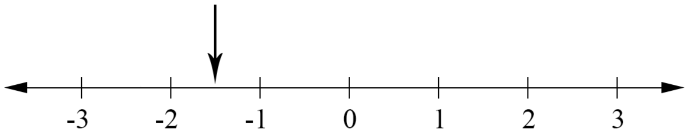
7. Number, NAP-85051



The arrow points to a position on the number line.

What number is at this position?

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8. Number, NAP-08573

Place	Population
Melbourne	3 879 261
Sandleford	2 321 898
Brownton	2 205 873
Chelsea	1 099 604
Priory	963 740
Hunter	308 997
Silby	444 820
Karuah	22 317

How many **more** people live in Sandleford than Silby and Karuah combined?

- 467 137
- 422 503
- 1 854 761
- 1 899 395

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9. Number, NAP-79132

Which of the following is equal to 32?

- $2^3 \times 2^2$
- $2^3 + 2^2$
- $3^2 + 2^2$
- $3^2 \times 2^2$

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10. Number, NAP-19142

The results of a 400 metres running race final was recorded in the table below.

1st place	52.26 seconds
2nd place	52.62 seconds
3rd place	?
4th place	53.19 seconds

The time of the runner who came in 3rd place could be

- 52.57 seconds.
- 52.71 seconds.
- 53.21 seconds.
- 53.94 seconds.

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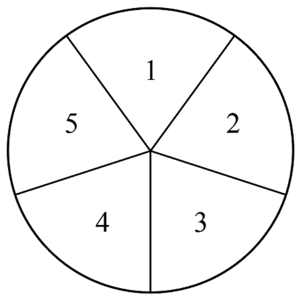
11. Number, NAP-42662

A circle is divided into 5 equal areas and labelled, as shown in the diagram below.



What percentage of the circle's area has been labelled with an odd number?

- 3%
- 15%
- 30%
- 40%
- 60%

12. Number, NAP-66184

Nigella placed a leg of lamb into her oven in the morning.

The 1st measurement of the lamb's temperature was $-6\,°C$.

Fifteen minutes later, a second temperature reading was taken, which measured $16\,°C$.

What was the change in temperature between the two measurements?

decrease of $22°C$ | decrease of $10°C$ | increase of $10°C$ | increase of $22°C$

13. Algebra, NAP-13229

Which expression is equal to $4a^4$?

- $8 \times a$
- $16 \times a$
- $4 \times a \times a \times a \times a$
- $4 + a + a + a + a$
- $4 \times 4 \times 4 \times 4 \times a \times a \times a \times a$

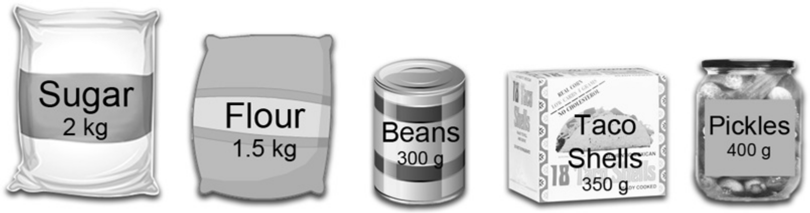
14. Number, NAP-72098

Which of the following changes is the smallest?

- $-8°\text{C}$ to $-4°\text{C}$
- $-5°\text{C}$ to $-2°\text{C}$
- $-2°\text{C}$ to $3°\text{C}$
- $3°\text{C}$ to $7°\text{C}$

15. Measurement, NAP-42665

Kim bought these 5 items.



The total mass of Kim's items is closest to

3 kg | 4 kg | 5 kg | 8 kg

16. Algebra, NAP-32096

If $x = 5$, what is the value of $\dfrac{3x}{2x-5}$?

2 | 3 | 4 | 15

17. Number, NAP-13223

Mark is booking a hotel room online that has a rating of 3.7 out of 5.

The ratings of the other 4 hotel rooms he considered are listed below.

- Hotel 57's rating 1.8
- Snuggy Hotel's rating 1.85
- Castle Hotel's rating 1.91
- Cross Hotel's rating 2.5

Which hotel has a rating that is exactly half that of the hotel Mark booked?

Hotel 57 | Snuggy Hotel | Castle Hotel | Cross Hotel

18. Number, NAP-73219

Megan has $\frac{1}{8}$ cup of sugar.

She needs $1\frac{1}{2}$ cups of sugar for a cake recipe she is using.

How much more sugar does Megan need for the recipe?

$1\frac{1}{8}$ cups $\quad 1\frac{1}{4}$ cups $\quad 1\frac{3}{8}$ cups $\quad 1\frac{5}{8}$ cups

19. Algebra, NAP-13256

Ryan and Oliver are buying goldfish for their fish tank.

Oliver buys 3 times the number of goldfish as Ryan does, plus another 5.

Let m be the number of goldfish that Ryan buys.

Which expression represents the number of goldfish Oliver buys?

$(3 \times m) + 5 \quad 3 + (5 \times m) \quad (m \div 3) - 5 \quad 3 - (m \div 5)$

20. Number, NAP-49699

What is the value of $\dfrac{74.2 + 13.56}{6.4 \times 2.1}$

$4.6 \quad 6.5 \quad 10.3 \quad 74.9$

21. Number, NAP-89706

$8 \times (4 + 3) \boxed{} 7 \times 5 + 3$

Which symbol ($<$, $=$, $>$) makes the number sentence correct?

$< \quad = \quad >$

22. Algebra, NAP-02655

$y = 4x^2 - 7$

What is the value of y when $x = 3.1$?

$5.4 \quad 17.8 \quad 31.44 \quad 146.76$

23. Number, NAP-02657

James weighed 80 kg when he was 18 years old.

On his 21st birthday, James weighs 5% more than when he was 18.

How much does he weigh now?

76 kg $\quad$ 84 kg $\quad$ 85 kg $\quad$ 120 kg

24. Number, NAP-07315

Which is the correct expression for 96.4851 rounded to 2 decimal places?

$96.00 \quad 96.48 \quad 96.49 \quad 96.50$

25. Number, NAP-60272

1102 is a tri-square number because

$$1^2 + 1^2 + 0^2 = 2$$

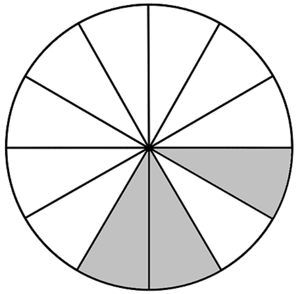
The next tri-square number after 1102 is

$1104 \quad 1113 \quad 1209 \quad 1126$

26. Number, NAP-36780

A birthday cake is cut into 12 equal pieces.

Only the shaded pieces below are left.



What percentage represents the amount of cake left?

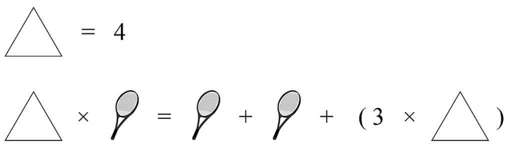
10% 20% 25% 33%

27. Algebra, NAP-79193

Which expression is equivalent to $6x + 12$?

$3(2x+4)$ $3(2x+9)$ $6(x+6)$ $6(x+12)$

28. Algebra, NAP-07314



What is the value of  ?

0 1 2 4 6

29. Number, NAP-08603

Jack was given a question without a calculator.

$$22 - 16 \times \sqrt{8.5 + 7.5} =$$

Which operation should Jack perform first?

$22 - 16$ $16 \times \sqrt{8.5}$ $\sqrt{8.5}$ $8.5 + 7.5$

30. Number, NAP-79164

Alison's petrol tank was empty.

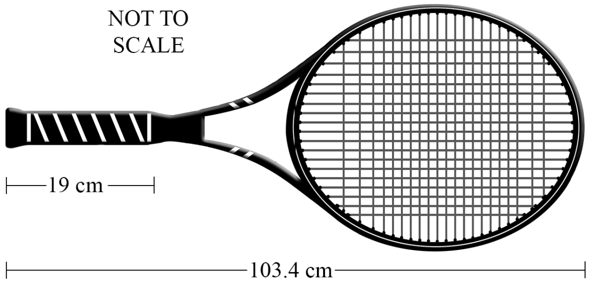
She then spent \$55 filling her tank up to half way at a cost of \$1.55 per litre.

Approximately how much petrol can Alison's petrol tank hold when it is full?

35 litres 65 litres 71 litres 85 litres

31. Number, NAP-49732

Ryan's tennis racquet is pictured below.



His handle is 19 cm in length.

Which of the following show the length of the handle as a percentage of the total length of the racquet?

11% 18% 19% 33%

32. ☒ Algebra, NAP-66243

Which of the following is always equal to $a - b$?

$-a-(-b)$ ○ $-a+b$ ○ $b-a$ ○ $-b+a$ ○

Worked Solutions

1. Algebra, NAP-14484

$$\begin{aligned} 5 \times ? + 15 &= 95 \\ 5 \times ? &= 80 \\ \therefore ? &= \frac{80}{5} \\ &= 16 \end{aligned}$$

2. Number, NAP-85047

$$\begin{aligned} \text{Distance} &= 14 \times 5.78 \\ &= 80.92 \text{ km} \end{aligned}$$

3. Number, NAP-08568

$$\text{Number of girls} = 288\,701 - 150\,125 = 138\,576$$

4. Number, NAP-66181

$$\text{Bags needed} = \frac{20}{4} = 5$$

5. Algebra, NAP-90964

$$\begin{aligned} p &= 7 \\ 2p &= 2 \times 7 \\ &= 14 \end{aligned}$$

6. Number, NAP-61527

$$\begin{aligned} \text{Fraction} &= \frac{\text{spaces filled}}{\text{total spaces}} \\ &= \frac{7}{9} \end{aligned}$$

7. Number, NAP-85051

-1.5

8. Number, NAP-08573

$$\begin{aligned}\text{Number of people} &= 2\,321\,898 - (444\,820 + 22\,317)\\ &= 1\,854\,761\end{aligned}$$

9. Number, NAP-79132

$$\begin{aligned}2^3 \times 2^2 &= 8 \times 4\\ &= 32\end{aligned}$$

10. Number, NAP-19142

52.71 is slower than 2nd place (52.62)

but faster than 4th place (53.19).

11. Number, NAP-42662

Since all areas are equal,

Percentage labelled with an odd number

$$\begin{aligned}&= \frac{3}{5}\\ &= 60\%\end{aligned}$$

12. Number, NAP-66184

$$\begin{aligned}\text{Change in temperature} &= 16 - (-6)\\ &= 22^\circ\text{C}\end{aligned}$$

$\therefore$ An increase of 22°C.

13. Algebra, NAP-13229

$4a^4 = 4 \times a \times a \times a \times a$

14. Number, NAP-72098

Consider each option:

$$\begin{aligned}-8 \text{ to } -4 &= 4^\circ \text{ change}\\ -5 \text{ to } -2 &= 3^\circ \text{ change}\\ -2 \text{ to } 3 &= 5^\circ \text{ change}\\ 3 \text{ to } 7 &= 4^\circ \text{ change}\end{aligned}$$

$\therefore$ -5° C to -2°C is the smallest change.

15. Measurement, NAP-42665

$$\begin{aligned}&2\text{ kg} + 1.5\text{ kg} + 300\text{ g} + 350\text{ g} + 400\text{ g}\\ &= 2\text{ kg} + 1.5\text{ kg} + 0.3\text{ kg} + 0.35\text{ kg} + 0.4\text{ kg}\\ &= 4.55\text{ kg}\\ &= 5\text{ kg} \ \ (\text{nearest kg})\end{aligned}$$

16. Algebra, NAP-32096

If $x = 5$,

$$\begin{aligned}\frac{3x}{2x-5} &= \frac{3 \times 5}{(2 \times 5) - 5}\\ &= \frac{15}{5}\\ &= 3\end{aligned}$$

17. Number, NAP-13223

$3.7 \div 2 = 1.85$

$\therefore$ Snuggy Hotel

18. Number, NAP-73219

$$\begin{aligned}\text{Sugar needed} &= 1\frac{1}{2} - \frac{1}{8}\\ &= 1\frac{3}{8} \text{ cups}\end{aligned}$$

19. Algebra, NAP-13256

Goldfish bought by Oliver

$$= (3 \times m) + 5$$

20. Number, NAP-49699

$$\frac{87.76}{13.44} = 6.529\ldots$$
$$= 6.5$$

21. Number, NAP-89706

$$8 \times (4 + 3) > 7 \times 5 + 3$$
$$8 \times 7 > 35 + 3$$
$$56 > 38$$

22. Algebra, NAP-02655

$$y = 4(3.1)^2 - 7$$
$$= 38.44 - 7$$
$$= 31.44$$

23. Number, NAP-02657

$$80 + (80 \times 5\%) = 80 + 4$$
$$= 84 \text{ kg}$$

24. Number, NAP-07315

96.49

25. Number, NAP-60272

$$1^2 + 1^2 + 1^2 = 1 + 1 + 1$$
$$= 3$$

$\therefore$ 1113 is the next tri-square number.

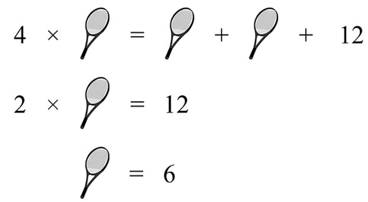
26. Number, NAP-36780

$$\text{Percentage} = \frac{3}{12}$$
$$= 0.25$$
$$= 25\%$$

27. Algebra, NAP-79193

$$3(2x + 4) = 6x + 12$$

28. Algebra, NAP-07314



29. Number, NAP-08603

The sum of the numbers under the square root symbol need to be calculated first.

$\therefore$ The first operation is $8.5 + 7.5$.

30. Number, NAP-79164

$$\text{Volume of half tank} = \frac{55}{1.55}$$
$$= 35.48\ldots$$

$\therefore$ Volume of full tank

$$= 2 \times 35.48\ldots$$
$$= 70.96\ldots$$
$$= 71 \text{ litres}$$

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31. Number, NAP-49732

Length of the handle as a percentage

$$= \frac{19}{103.4} \times 100$$
$$= 18.4\ldots\ \%$$

$\therefore$ 18% is the closest option.

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32. Algebra, NAP-66243

$a - b = \ -b + a$
